Appendix A - Geographe Marine Park Natural Values Report

South-west Marine Parks Network

Table of contents

Natural Values

Understanding of natural values

Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone, SZ = Sanctuary Zone, GUZ = General Use Zone.

					Habitat	Fish	
Date	**Campaign name**	**Method**	**Areas**	**Samples**	**Count**	**Count**	**Length**
Dec 2014	2014-12 Geographe-bay stereo-BRUVs	BRUV	Coastal waters, GUZ, HPZ, MUZ, NPZ, SPZ, SZ	324	320	297	238
Mar-Apr 2024	2024-04 Geographe stereo-BRUVs	BRUV	Coastal waters, GUZ, HPZ, MUZ, NPZ, SPZ, SZ	261	260	259	246
Apr 2024	2024-04 Geographe BOSS	BOSS	GUZ, HPZ, MUZ, NPZ	121	121	-	-

Benthic ecosystem extent

Observations

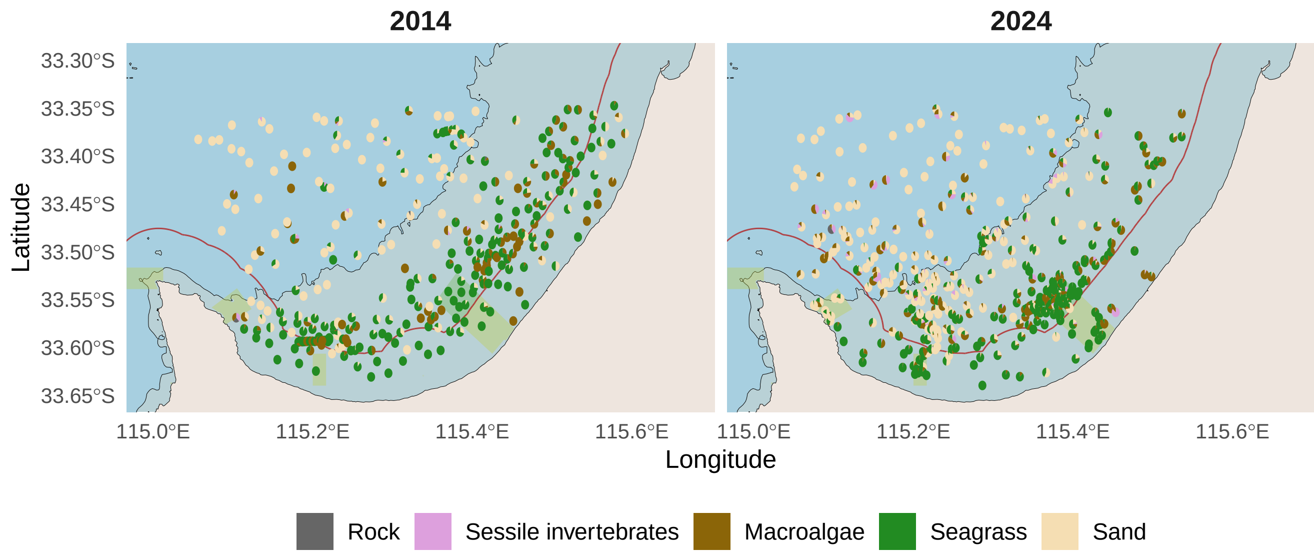

Figure 1: Habitat classes observed in stereo-BRUV imagery. Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

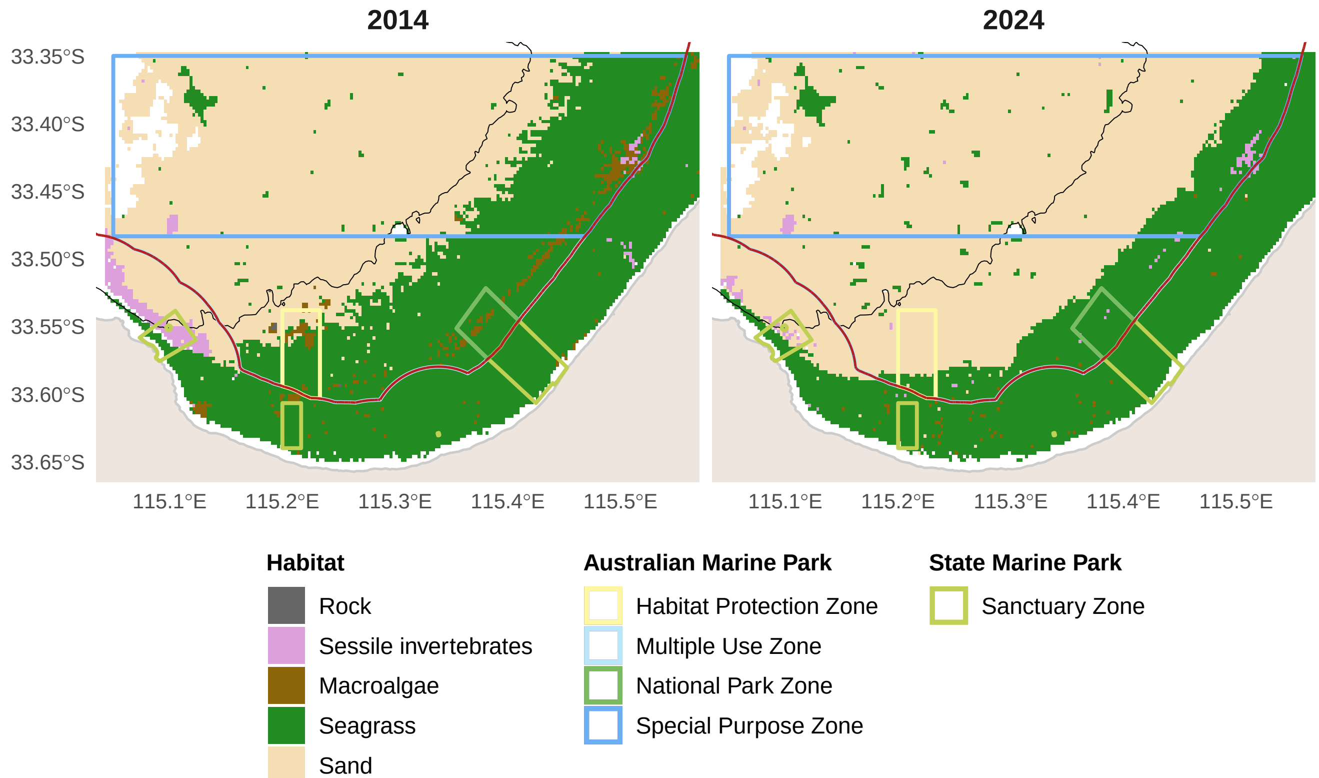

Figure 2: Predicted dominant habitat type from stereo-BRUV groundtruthing.The green polygon represents the extent of no-take National Park Zones and State no-take sanctuary zones. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

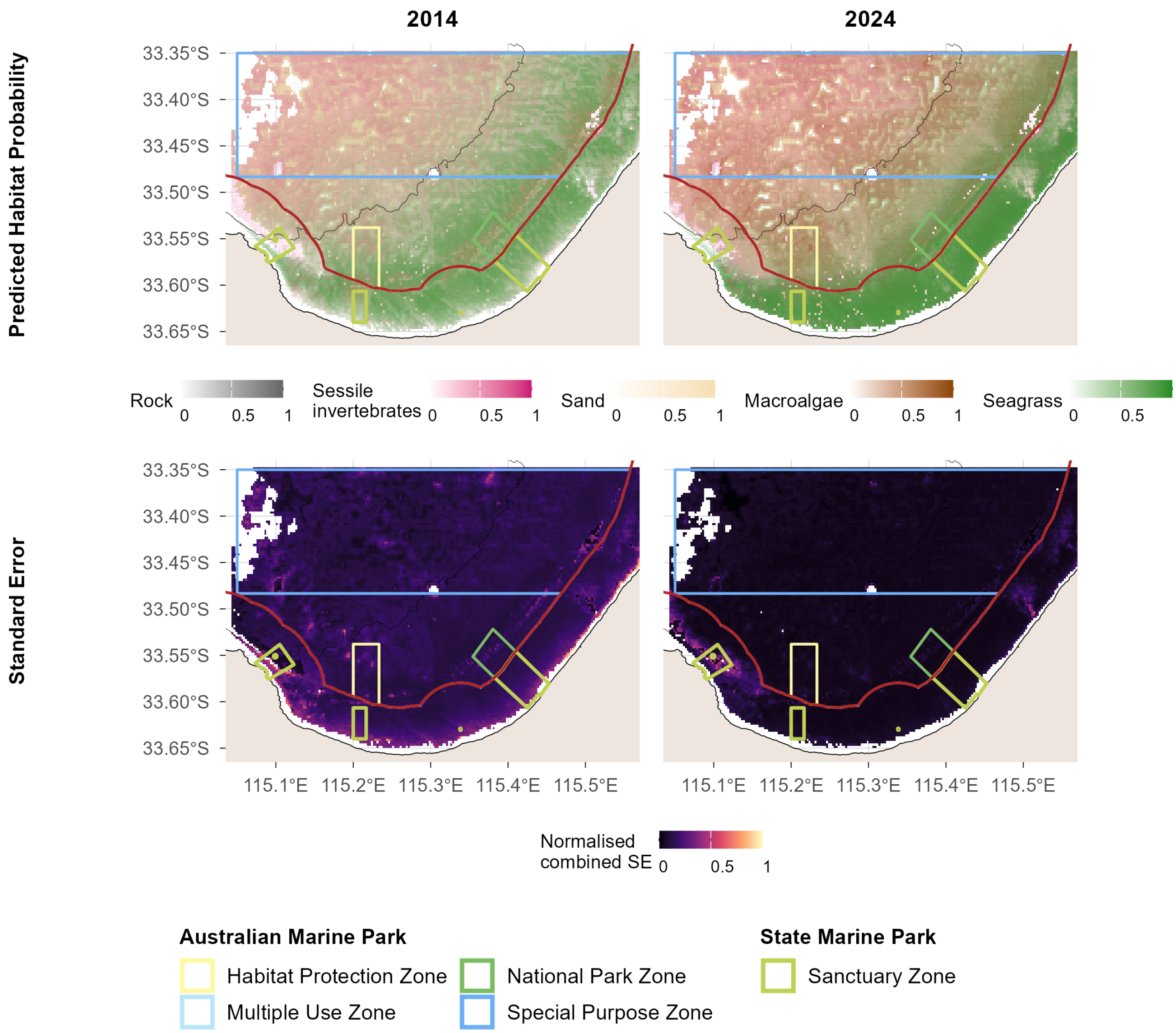

Figure 3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. The green polygon represents the extent of no-take National Park Zones. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Functional reef distribution and probability

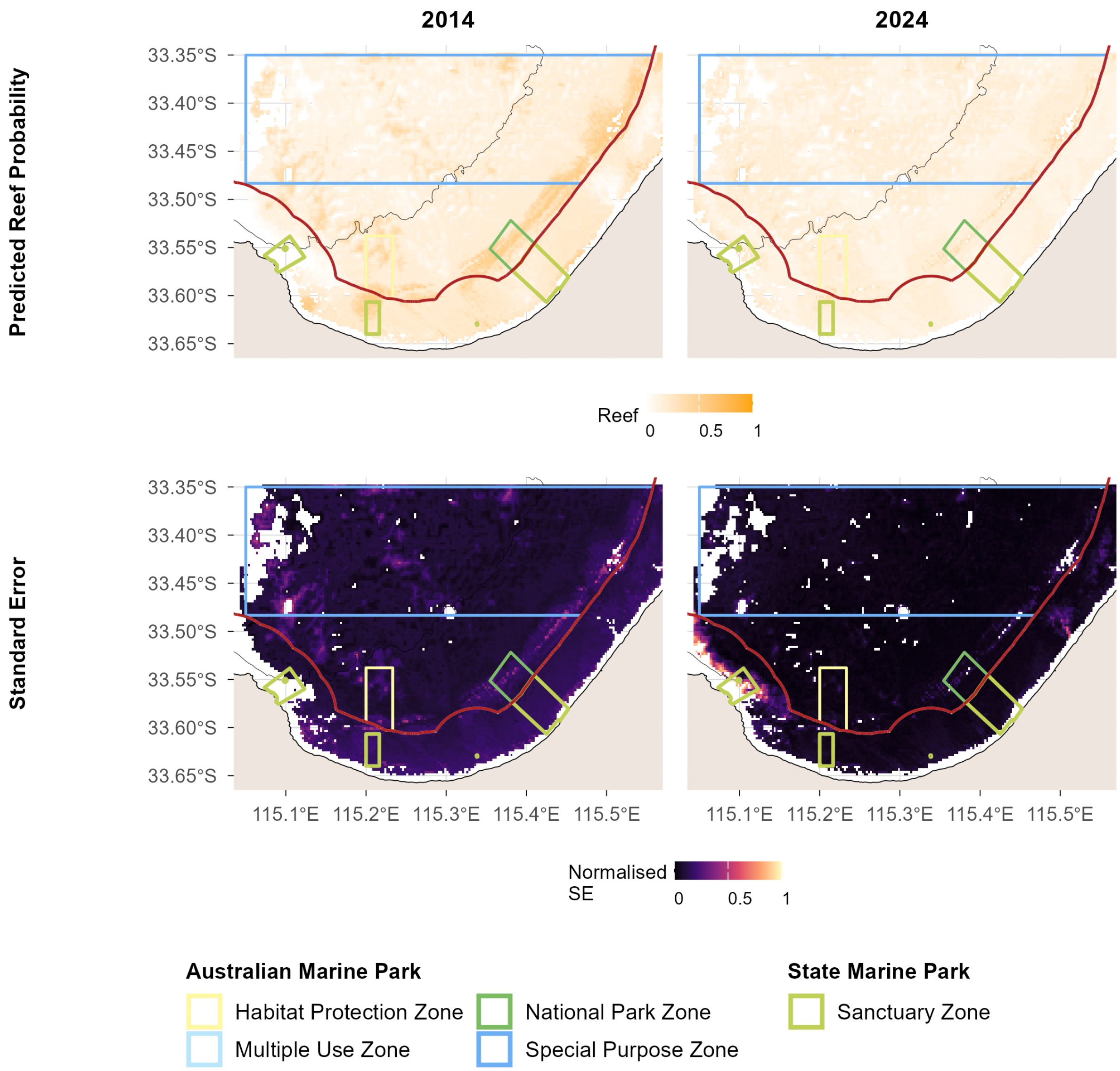

Figure 4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing.The green polygon represents the extent of no-take National Park Zones. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Seagrass

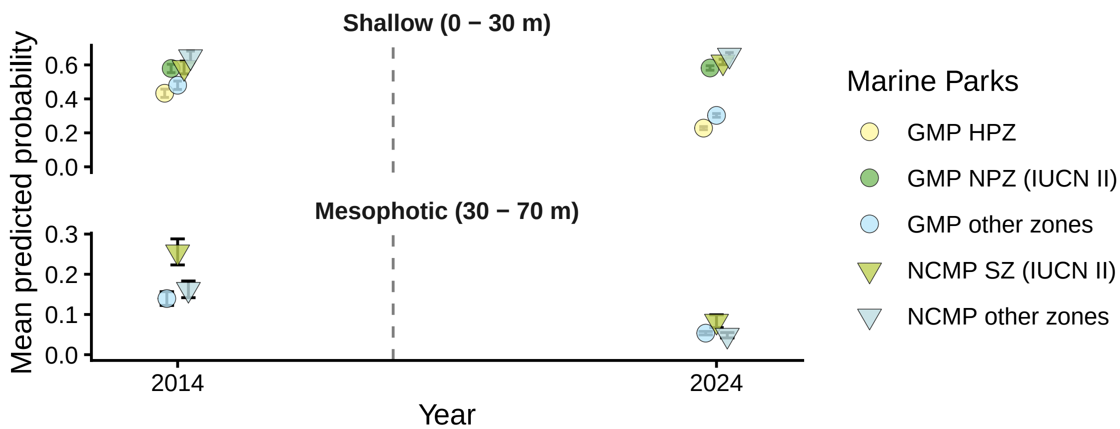

Figure 5.1: Seagrass: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

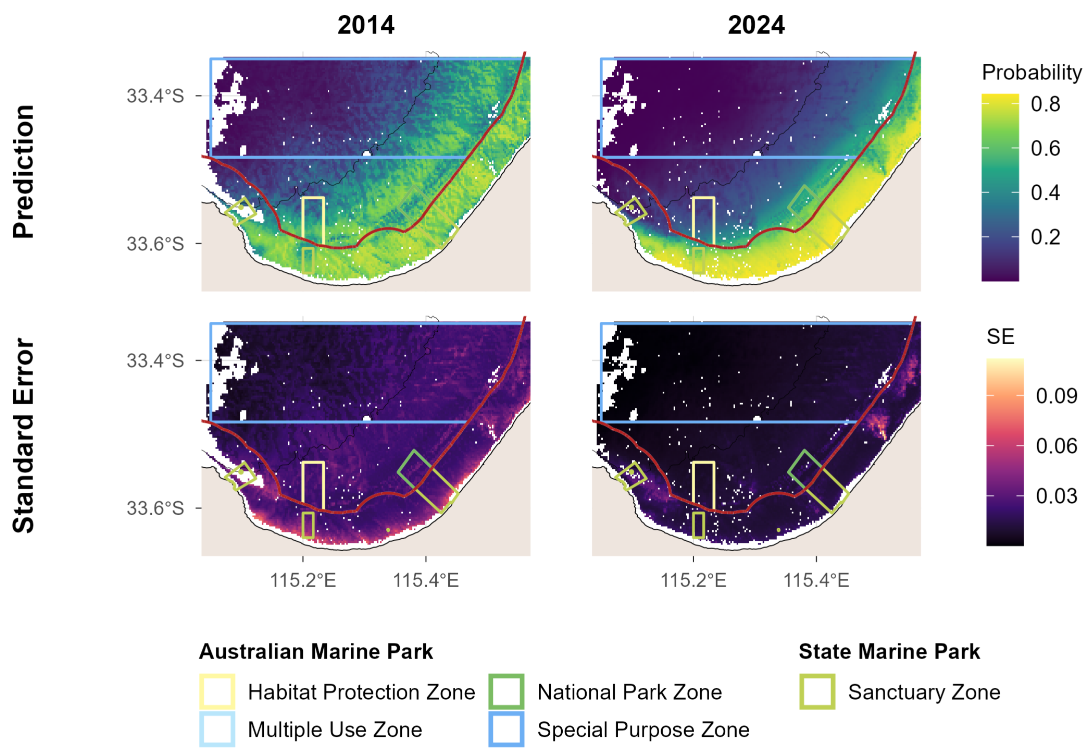

Figure 5.2: Seagrass: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Macroalgae

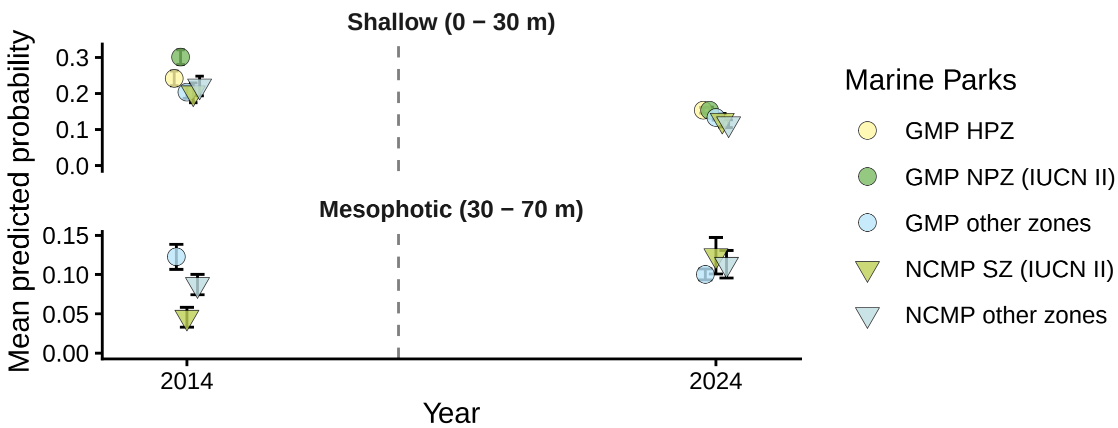

Figure 6.1: Macroalgae: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

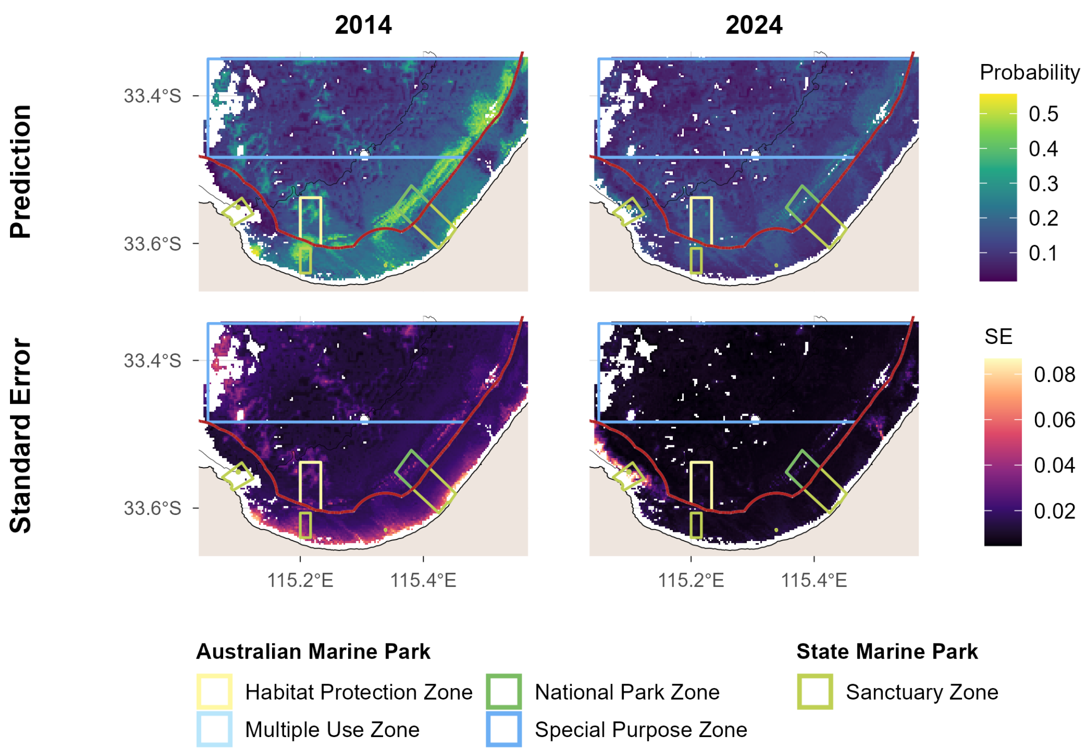

Figure 6.2: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

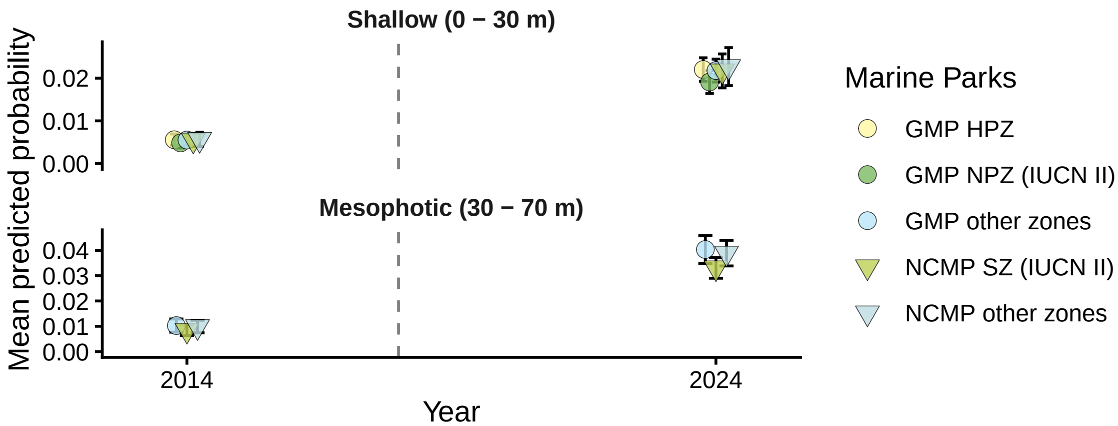

Figure 7.1: Sessile invertebrates: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

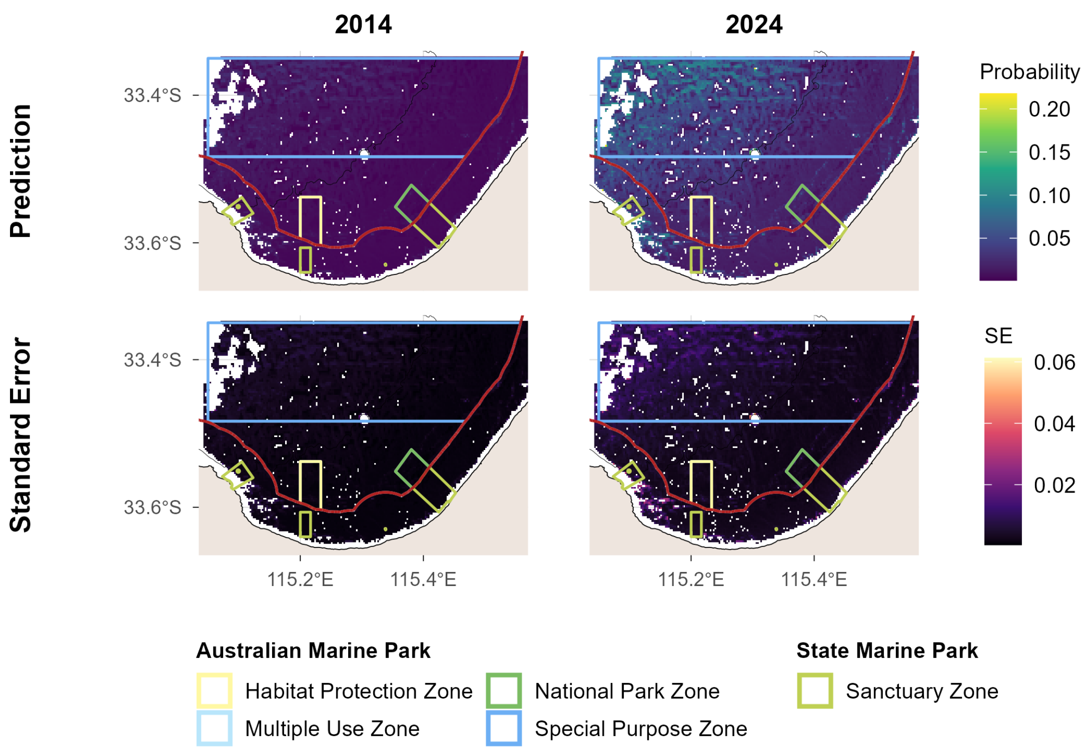

Figure 7.2: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of demersal fish natural values

Survey effort

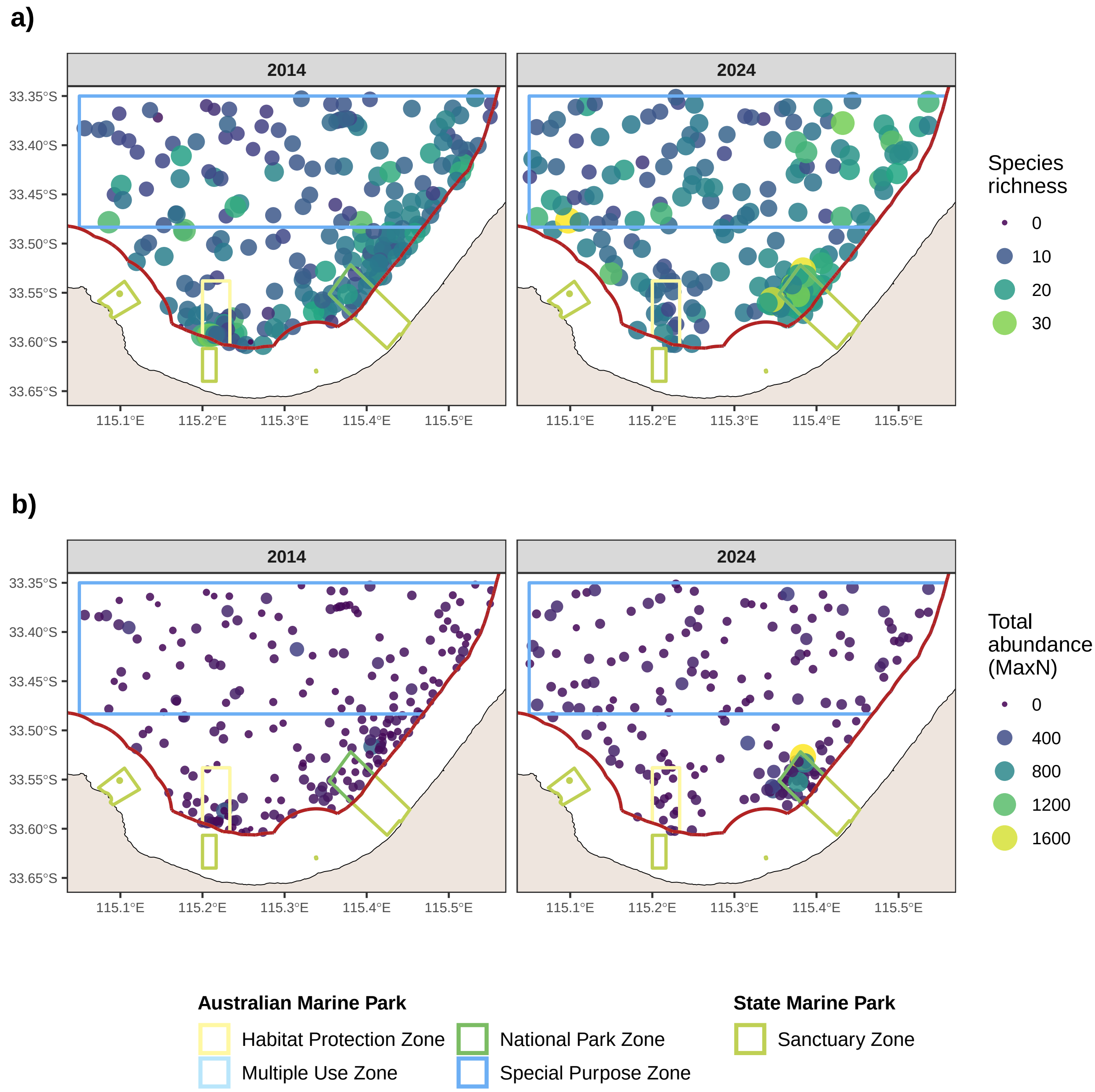

Figure 8: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

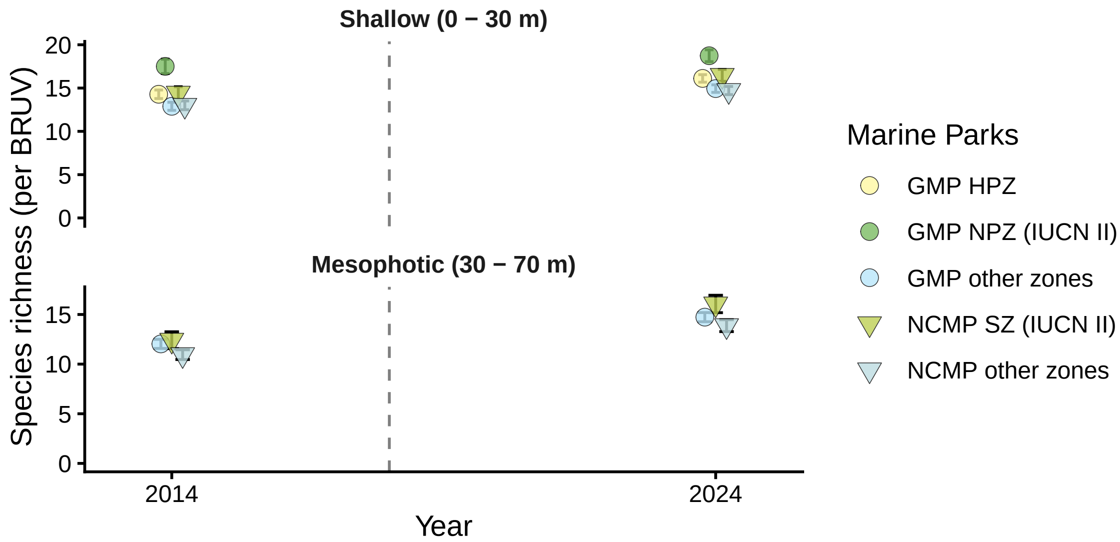

Figure 9.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

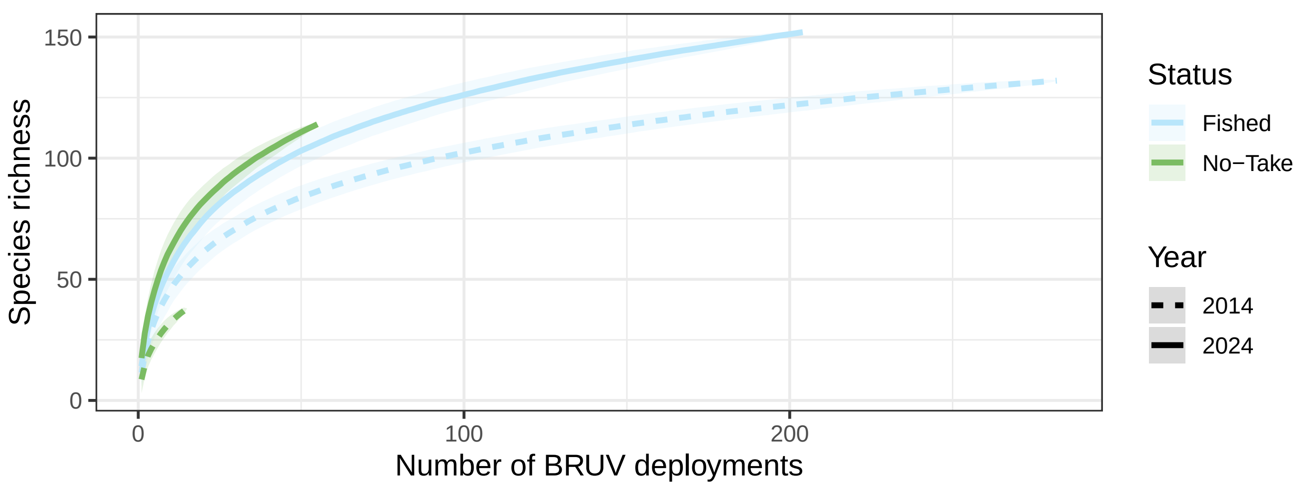

Figure 9.2: Demersal fish - Species richness: species accumulation curve by zone type from stereo-BRUV groundtruthing.

Species richness

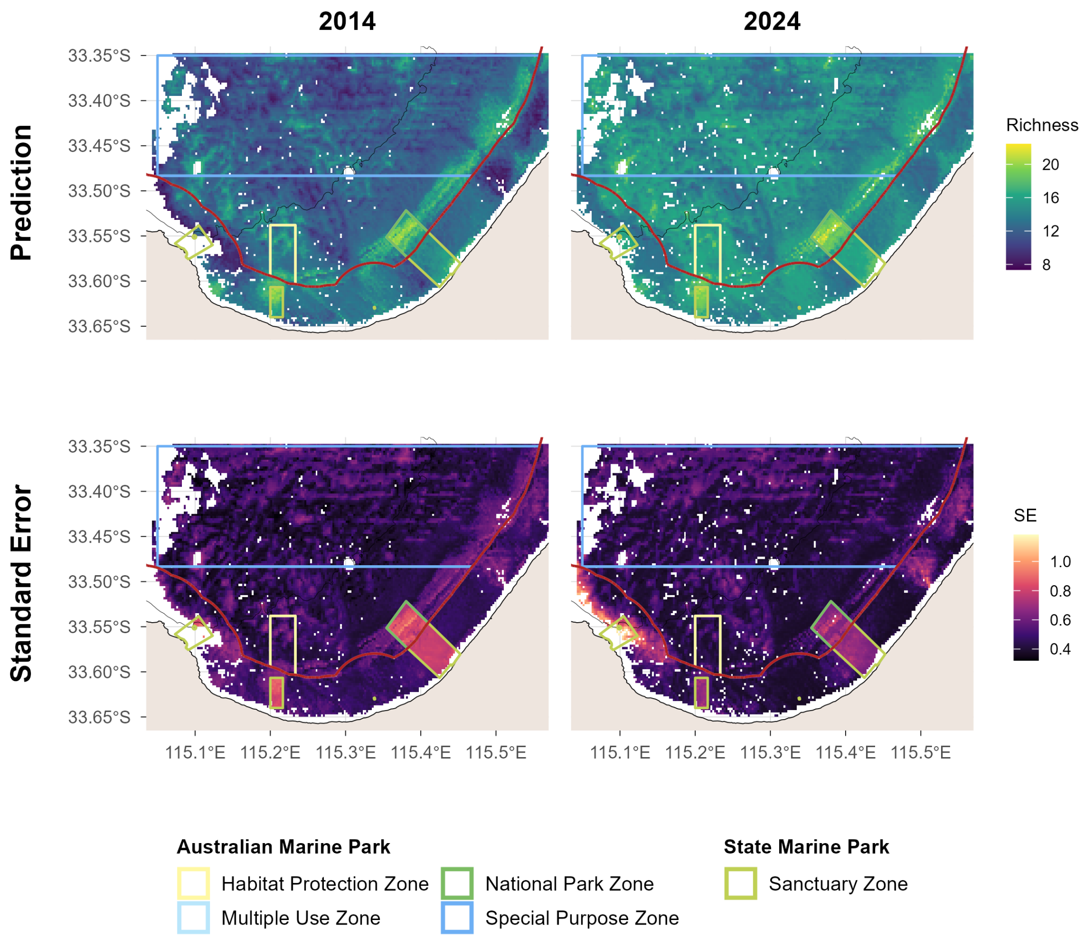

Figure 9.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

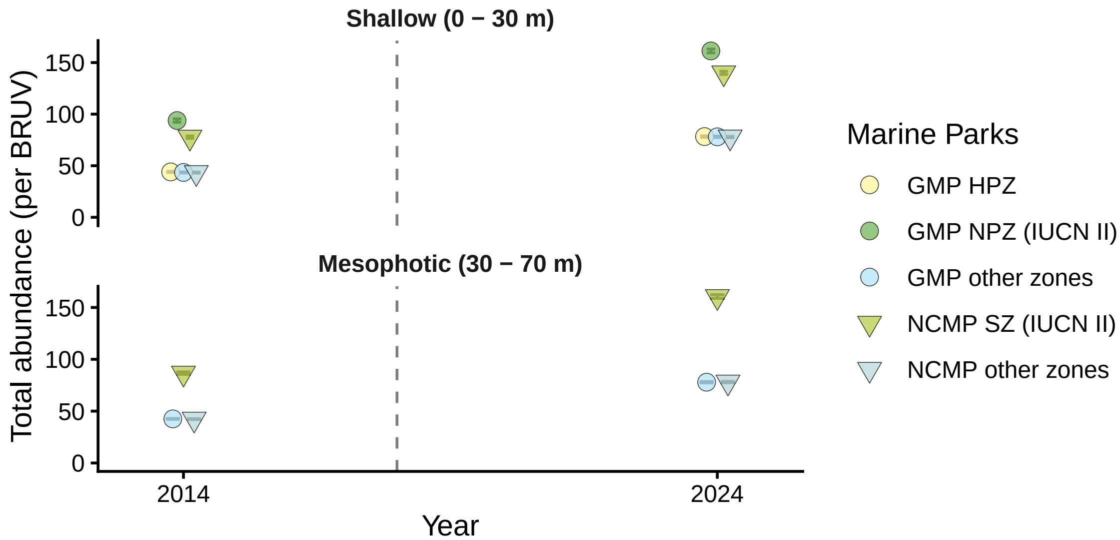

Figure 10.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

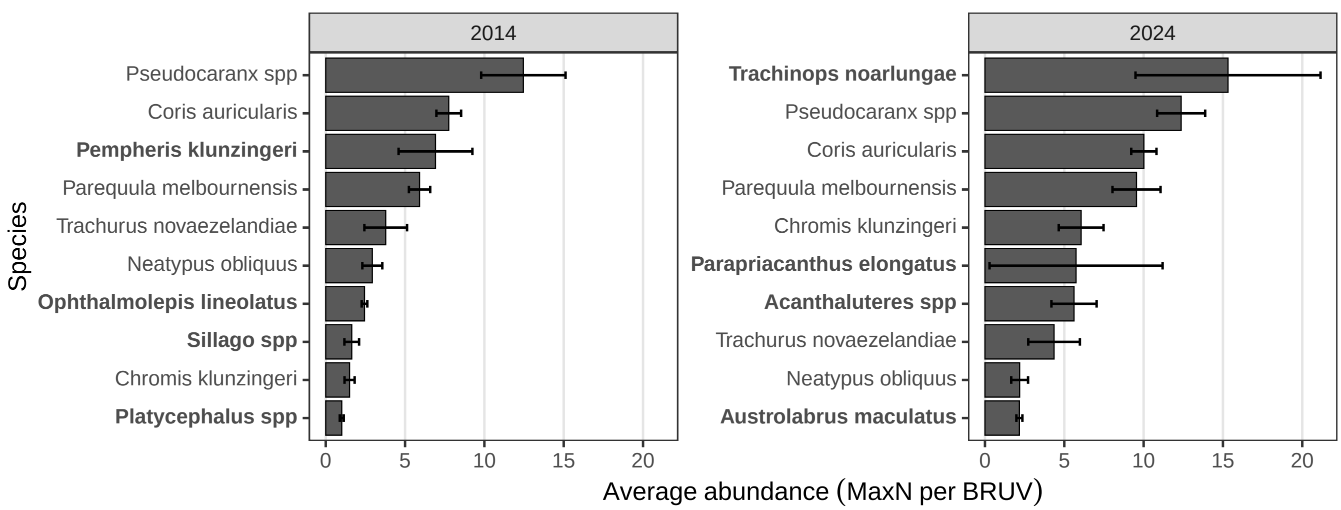

Figure 10.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 most abundant species by year.

Total abundance

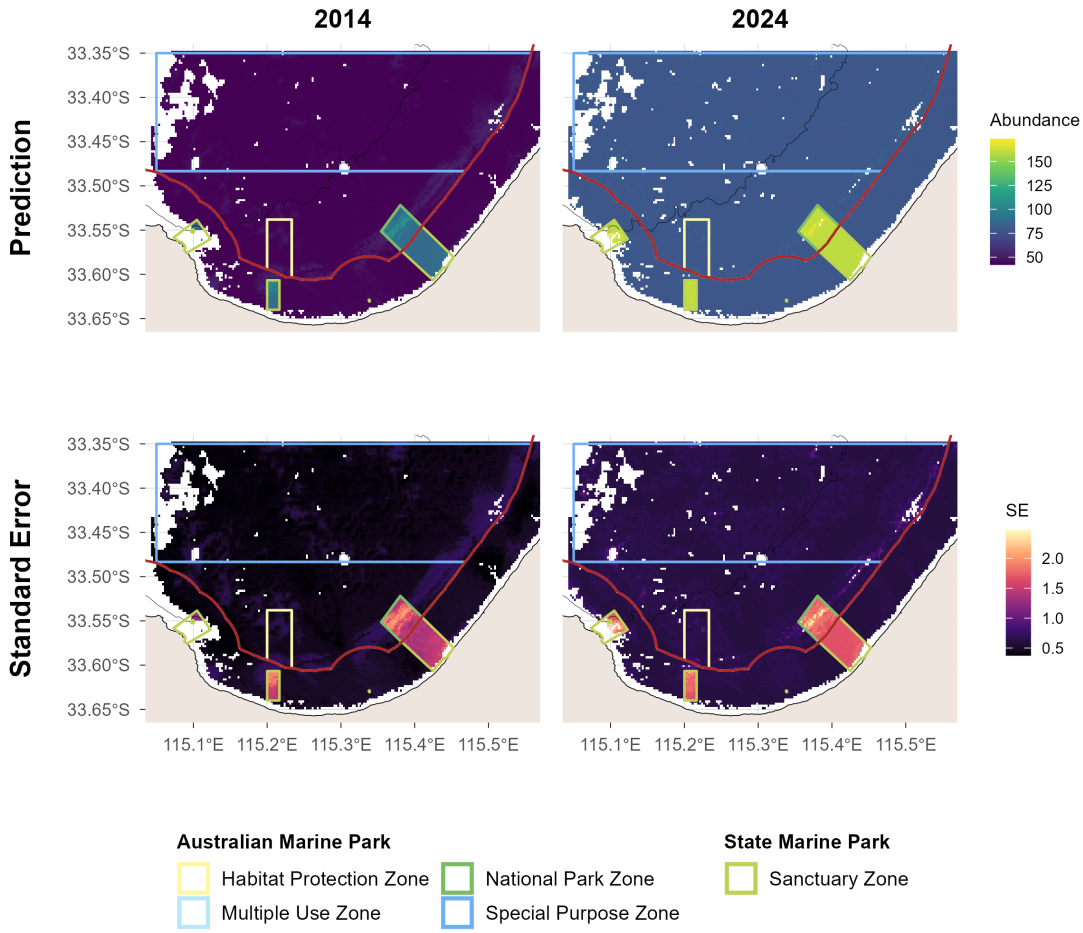

Figure 10.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

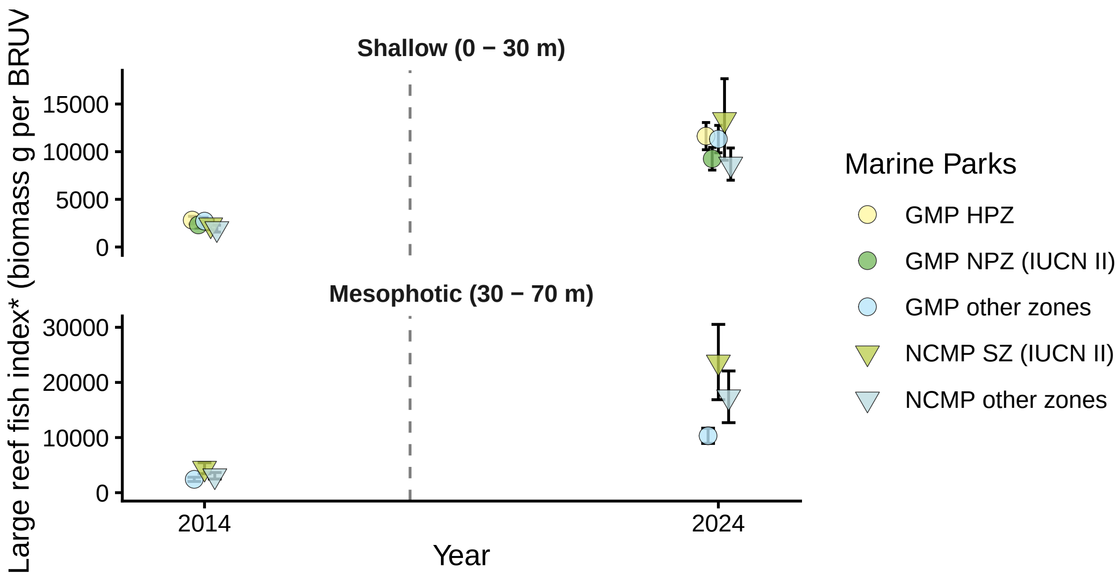

Figure 11.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

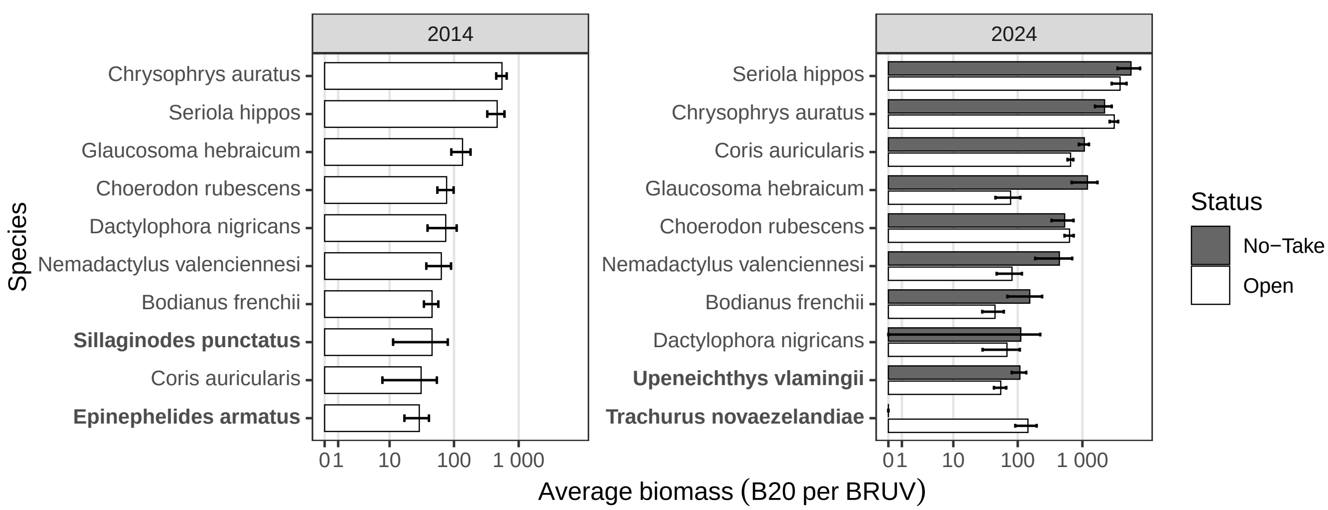

Figure 11.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species by status (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 greatest biomass species by year.

Large Reef Fish Index*

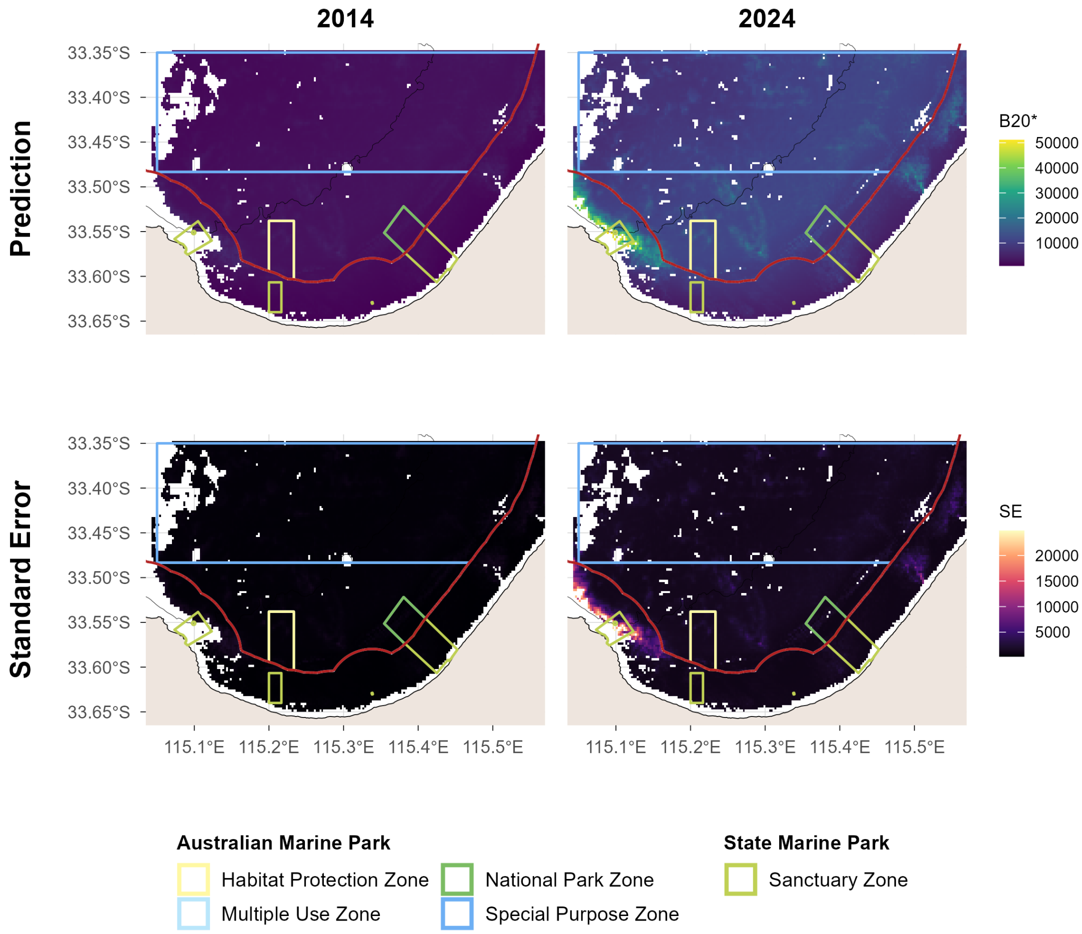

Figure 11.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

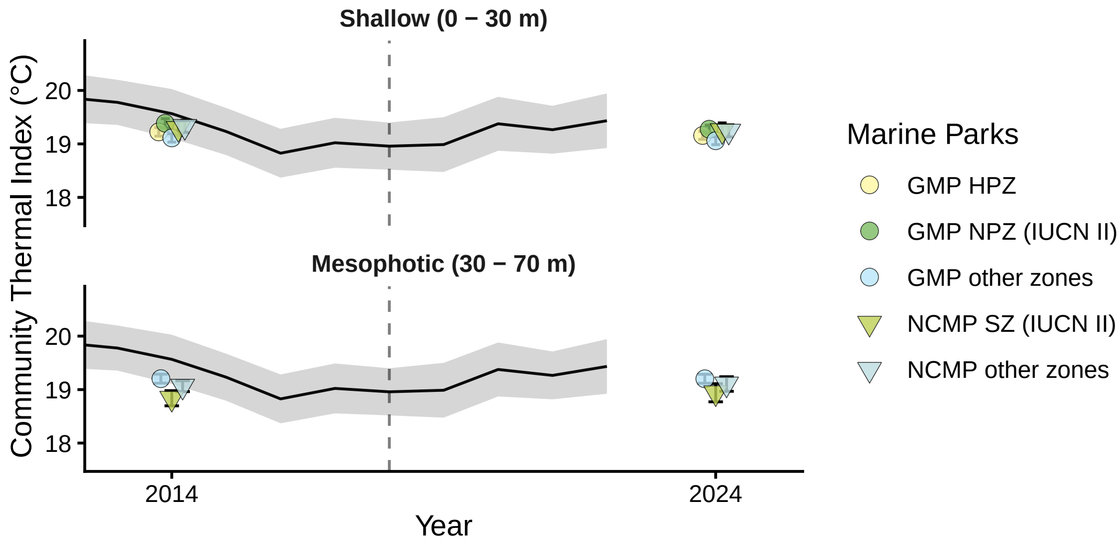

Figure 12.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

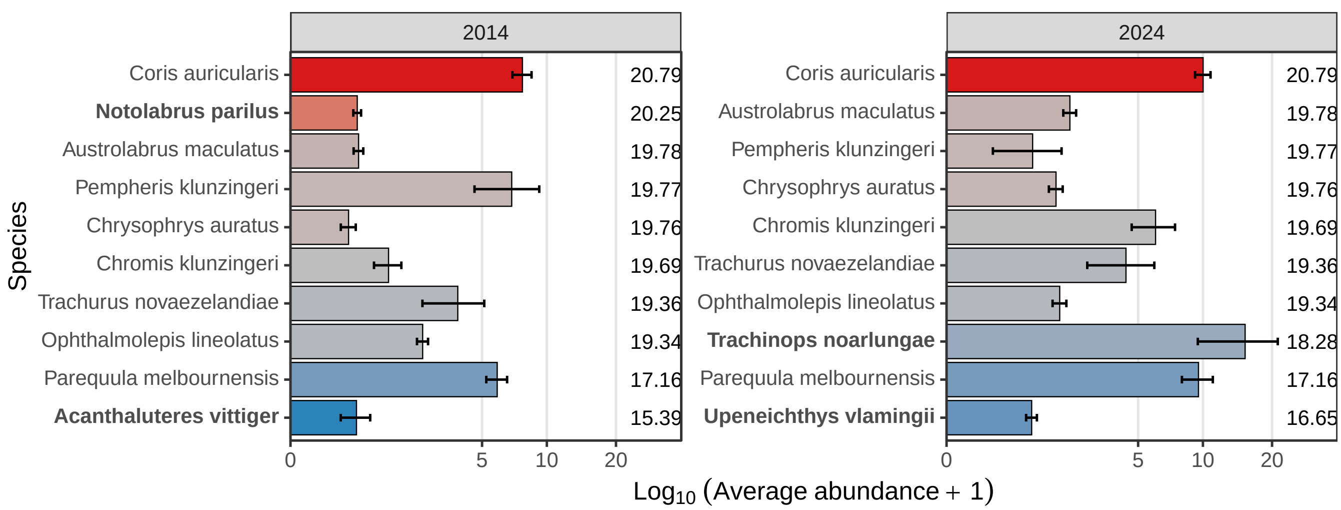

Figure 12.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 most abundant species by year.

Reef fish thermal index

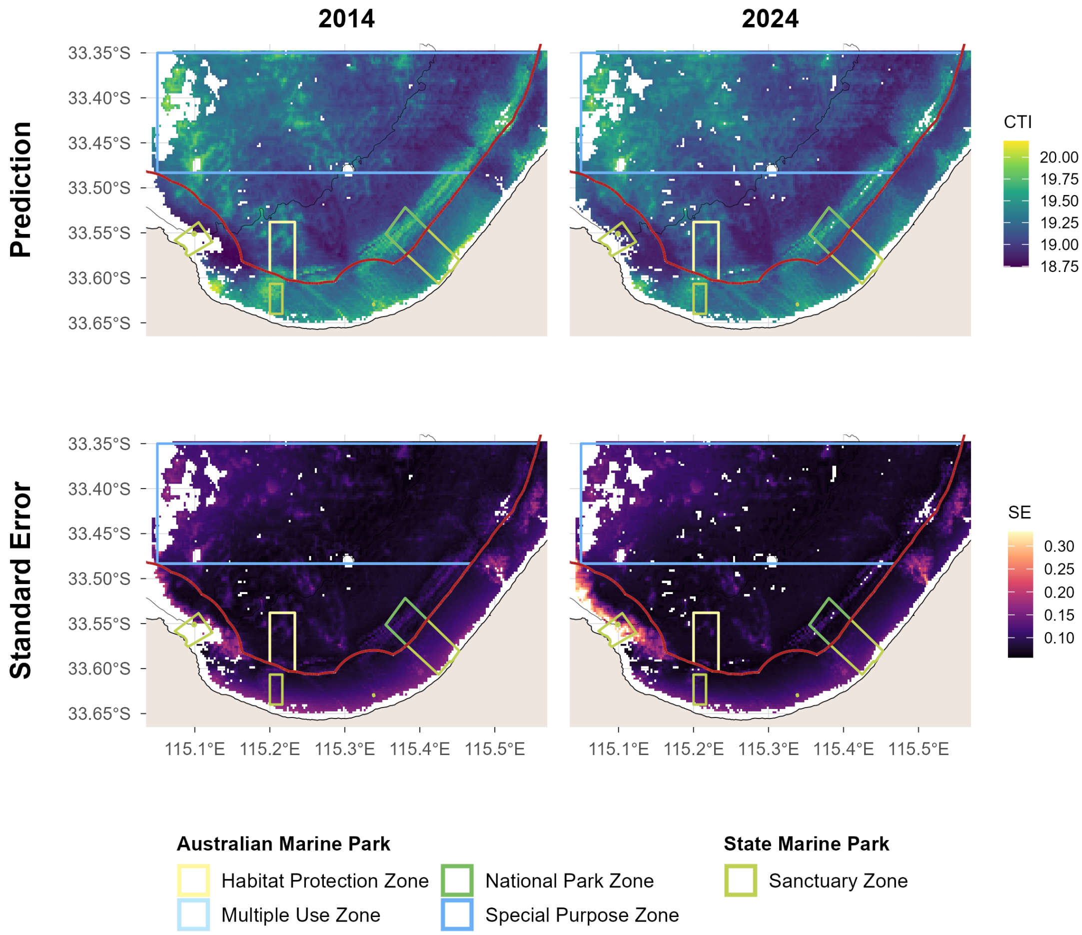

Figure 12.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.